

16 Flossing Effects on Range of Motion and Performance

Floss bands were initially developed by Kelly Starrett (physical therapist) (15). These rubber bands are wrapped around joints or muscles while exercising or stretching (20) (Figure 16.1 and Figure 16.2). Floss bands that can be applied before or after activity are a relatively recent tool purported to ameliorate joint range of motion (ROM) (3, 4, 13, 20), reduce pain (6), prevent sports injuries (25) and improve sports performance (20).

Floss Band Effects on Range of Motion

A number of original research studies have reported ROM improvements with the quadriceps and hamstrings (2, 23, 30), dorsiflexion (5), and plantar flexion ROM (3, 4). Flossing provided greater improvements than dynamic stretching in supine leg raise scores and maximal eccentric knee



Figure 16.1 Floss band on calf muscles

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Figure 16.2 Floss band on upper arm

extension contraction force (12). The combination of flossing and active isolated stretching of the hamstrings (three sets of 10 repetitions) tended to be better than stretching without flossing on the sit-and-reach (hamstrings and lower back) test (17). However as with all science, there is always conflicting research with reports of no improvements in plantar flexion or ankle ROM (5, 11), sit and reach scores (9), or the active straight leg raise test (29). So, this is why systematic reviews and meta-analyses were developed. With a meta-analysis we can quantify an overall score based on the literature and the number of participants (more subjects normally suggest a statistically higher-powered study and thus these studies are given more weight in the meta-analysis). Thus, a review of 24 studies by the rising Austrian research star (and former professional soccer/football goalie) Andreas Konrad and colleagues revealed that a single floss band treatment provides small to moderate magnitude increases in ROM. Further sub-group analysis also found weak evidence that flossing during a warm-up may provide greater ROM improvements than stretching, but there was no superior effect when compared to foam rolling (20).

Flossing Effects on Performance

A number of original studies have shown increases in quadriceps isometric force (2, 29), single-leg jump performance (4), countermovement jump and sprint (3, 18, 24), dorsiflexion power (5), agility for one hour after flossing (10), single-leg hop distance and landing stabilization performance in women

for up to 20 minutes (30). It may also have no negative effect (30) or actually improve dynamic balance without impairing knee proprioception (2). Meanwhile, to the contrary, there are also reports of floss-induced impairments in knee flexion and extension peak torque, work, and power as well as no significant changes in time to peak torque or flexors–extensors ratio (8) or jump power (23). The Konrad meta-analytical review indicated that flossing can positively affect jumping and strength performance (small to moderate magnitude effects) (20).

Mechanisms

Most floss bands are applied tight enough to restrict blood flow. Hence, some of the mechanisms underlying flossing are reported to be reperfusion, fascial shearing, and blood occlusion (12). The floss-induced ischemia (from occlusion) and subsequent blood flow reperfusion (hyperemia) following its removal may enhance metabolic activity and elevate growth hormone and catecholamine levels (30). Flossing may also increase the pain threshold (e.g., gate control theory, and diffuse noxious inhibitory control release of endorphins and enkephalins) (16, 28), which is also a commonly attributed mechanism for augmenting ROM with stretching (1, 21, 22). The Konrad et al. review (20) suggested that a possible mechanism for floss band-induced increases in ROM or performance is more likely associated with alterations in neuromuscular function (e.g., electromyography, EMG) than muscle mechanical properties (e.g., stiffness). However, contradictory results from other studies indicated that Achilles tendon and gastrocnemius stiffness were reduced for 60 minutes after flossing (18), which may have contributed to improved sprint and jump height, while Pasurka et al. (26) found decreases in ankle capsule and talofibular ligament stiffness as well as reactive hyperemia. An original piece of research from the Konrad group found no floss-induced increases in knee joint ROM but an increase in knee extensors maximum voluntary isometric contraction force (19). As there were no changes in passive resistive torque or EMG, they proposed that there might be an enhancement of growth hormone and norepinephrine after compression release. Another study found that tissue flossing decreased the soleus H-reflex (afferent excitability of the motoneurons) for ten minutes, but this depression did not result in greater ankle joint flexibility or plantar flexor strength (11).

Flossing Effects on Pain

According to the Konrad review (20), there is weak evidence that flossing can relieve pain when treating specific diseases or accelerating recovery after exercise. However, as expected the findings are not unanimous as Gorny and Stoggl (7) reported that flossing did not reduce DOMS symptoms after repetitive leg presses, whereas Prill et al. (27) reported that flossing diminished DOMS (decreased inflammation) by a small magnitude for 48 hours post-

exercise. Tissue flossing provided large magnitude improvements with Pain Visual Analog Scale, Short Form McGill Pain Questionnaire II, pain-pressure threshold, hand grip dynamometry and a Likert scale for movement ability in resistance trained individuals with a history of musculoskeletal pain for greater than one month (14).

Summary

Whereas there are always conflicting studies, the overall trend highlights that flossing can increase ROM, improve static (isometric) force, power, and perhaps balance, with some evidence of pain relief. These benefits may stem from a variety of mechanisms related to occlusion-induced reperfusion enhancing metabolic activity and elevating growth hormone and catecholamine levels. In addition, flossing may reduce musculotendinous and ligamentous stiffness and increase pain thresholds.

References

1. Behm, D.G., Blazevich, A.J., Kay, A.D., and McHugh, M. Acute effects of muscle stretching on physical performance, range of motion, and injury incidence in healthy active individuals: a systematic review. *Appl Physiol Nutr Metab* 41: 1–11, 2016.
2. Chang, N.-J., Hung, W.-C., Lee, C.-L., Chang, W.-D., and Wu, B.-H. Effects of a Single Session of Floss Band Intervention on Flexibility of Thigh, Knee Joint Proprioception, Muscle Force Output, and Dynamic Balance in Young Adults. *Applied Sciences* 11: 12052, 2021.
3. Driller, M., Mackay, K., Mills, B., and Tavares, F. Tissue flossing on ankle range of motion, jump and sprint performance: A follow-up study. *Phys Ther Sport* 28: 29–33, 2017.
4. Driller, M.W. and Overmayer, R.G. The effects of tissue flossing on ankle range of motion and jump performance. *Phys Ther Sport* 25: 20–24, 2017.
5. Galis, J. and Cooper, D.J. Application of a Floss Band at Differing Pressure Levels: Effects at the Ankle Joint. *J Strength Cond Res* 36: 2454–2460, 2022.
6. Garcia-Luna, M.A., Cortello-Tormo, J.M., Gonzalez-Martinez, J., and Garcia-Jaen, M. The effects of tissue flossing on perceived knee pain and jump performance: A pilot study. *International Journal of Human Movement and Sport Science* 8: 63–68, 2020.
7. Gorny, V.S. and Stöggel, T. Tissue flossing as a recovery tool for the lower extremity after strength endurance intervals. *Sportverl Sportschad* 32: 55–60, 2018.
8. Hadamus, A., Jankowski, T., Wiaderna, K., Bugalska, A., Marszałek, W., Blazkiewicz, M., and Białoszewski, D. Effectiveness of Warm-Up Exercises with Tissue Flossing in Increasing Muscle Strength. *J Clin Med* 11, 2022.
9. Hadamus, A., Kowalska, M., Kedra, M., Wiaderna, K., and Białoszewski, D. Effect of hamstring tissue flossing during warm-up on sit and reach performance. *J Sports Med Phys Fitness* 62: 51–55, 2022.
10. Huang, Y.S., Lee, C.L., Chang, W.D., and Chang, N.J. Comparing the effectiveness of tissue flossing applied to ankle joint versus calf muscle on exercise performance in female adults: An observational, randomized crossover trial. *J Bodyw Mov Ther* 36: 171–177, 2023.

11. Kalc, M., Mikl, S., Zoks, F., Vogrin, M., and Stoggl, T. Effects of Different Tissue Flossing Applications on Range of Motion, Maximum Voluntary Contraction, and H-Reflex in Young Martial Arts Fighters. *Front Physiol* 12: 752641, 2021.
12. Kaneda, H., Takahira, N., Tsuda, K., Tozaki, K., Kudo, S., Takahashi, Y., Sasaki, S., and Kenmoku, T. Effects of Tissue Flossing and Dynamic Stretching on Hamstring Muscles Function. *J Sports Sci Med* 19: 681–689, 2020.
13. Kaneda, H., Naonobu, T., Kouji, T., Kiyoshi, T., Kenta, S., Sho, K., Yoshiki, T., Shuichi, S., Kensuke, F., and Tomonori, K. The effects of tissue flossing and static stretching on gastrocnemius exertion and flexibility. *Isokinetics and Exercise Science* 28: 205–213, 2020.
14. Kelly, C.F., Oliveri, Z., Saladino, J., Senatore, J., Kamat, A., Zarour, J., and Douris, P.C. The Acute Effect of Tissue Flossing on Pain, Function, and Perception of Movement: A Pilot Study. *Int J Exerc Sci* 16: 855–865, 2023.
15. Kiefer, B., Lemarr, K.E., Enriquez, C., and Tivener, K.A. A pilot study: Psychological effects of the voodoo floss band on glenohumeral flexibility. *International Journal of Athletic Therapy and Training* 22: 1–16, 2021.
16. Kielur, D.S. and Powden, C.J. Changes of ankle dorsiflexion using compression tissue flossing: A systematic review and meta-analysis. *Journal of Sport Rehabilitation* 30: 306–314, 2021.
17. Kitsuksan, T. and Earde, P. The immediate effects of tissue flossing during active isolated stretching on hamstring flexibility in young healthy individuals. *Physiotherapy Quarterly* 30: 61–67, 2021.
18. Klich, S., Smoter, M., Michalik, K., Bogdanski, B., Valera Calero, J.A., Manuel Clemente, F., Makar, P., and Mroczek, D. Foam rolling and tissue flossing of the triceps surae muscle: an acute effect on Achilles tendon stiffness, jump height and sprint performance – a randomized controlled trial. *Res Sports Med*: 1–14, 2022.
19. Konrad, A., Bernsteiner, D., Budini, F., Reiner, M.M., Glashuttner, C., Berger, C., and Tilp, M. Tissue flossing of the thigh increases isometric strength acutely but has no effects on flexibility or jump height. *Eur J Sport Sci* 21: 1648–1658, 2021.
20. Konrad, A., Mocnik, R., and Nakamura, M. Effects of Tissue Flossing on the Healthy and Impaired Musculoskeletal System: A Scoping Review. *Front Physiol* 12: 666129, 2021.
21. Magnusson, S.P. and Renstrom, P. The European College of Sports Sciences Position statement: The role of stretching exercises in sports. *European Journal of Sport Science* 6: 87–91, 2006.
22. Magnusson, S.P., Simonsen, E.B., Aagaard, P., Sorensen, H., and Kjaer, M.A. A mechanism for altered flexibility in human skeletal muscle. *Journal of Physiology* 497 (Pt 1): 291–298, 1996.
23. Maust, Z., Bradney, D., Collins, S.M., Wesley, C., and Bowman, T.G. The Effects of Soft Tissue Flossing on Hamstring Range of Motion and Lower Extremity Power. *Int J Sports Phys Ther* 16: 689–694, 2021.
24. Mills, B., Mayo, B., Tavares, F., and Driller, M. The Effect of Tissue Flossing on Ankle Range of Motion, Jump, and Sprint Performance in Elite Rugby Union Athletes. *J Sport Rehabil* 29: 282–286, 2020.
25. Mohamed, E.E., Useh, U., and Mtshali, B.F. Q-angle, pelvic width, and intercondylar notch width as predictors of knee injuries in women soccer players in South Africa. *African Health Science* 12: 174–180, 2012.
26. Pasurka, M., Lutter, C., Hoppe, M.W., Heiss, R., Gaulrapp, H., Ernstberger, A., Engelhardt, M., Grim, C., Forst, R., and Hotfiel, T. Ankle flossing alters

- periarticular stiffness and arterial blood flow in asymptomatic athletes. *J Sports Med Phys Fitness* 60: 1453–1461, 2020.
27. Prill, R., Schulz, R., and Michel, S. Tissue flossing: a new short-term compression therapy for reducing exercise-induced delayed-onset muscle soreness. A randomized, controlled and double-blind pilot crossover trial. *J Sports Med Phys Fitness* 59: 861–867, 2019.
 28. Stevenson, P.J., Stevenson, R.K., and Duarte, K.W. Acute effects of the voodoo flossing band on ankle range of motion. *Journal of Medicine, and Biomedical Applied Science* 7: 244–253, 2019.
 29. Vogrin, M., Kalc, M., and Ličen, T. Acute Effects of Tissue Flossing Around the Upper Thigh on Neuromuscular Performance: A Study Using Different Degrees of Wrapping Pressure. *Journal of Sport Rehabilitation* 30: 601–608, 2021.
 30. Wu, S.Y., Tsai, Y.H., Wang, Y.T., Chang, W.D., Lee, C.L., Kuo, C.A., and Chang, N.J. Acute Effects of Tissue Flossing Coupled with Functional Movements on Knee Range of Motion, Static Balance, in Single-Leg Hop Distance, and Landing Stabilization Performance in Female College Students. *Int J Environ Res Public Health* 19, 2022.